Alibi Shokputov

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EDUCATION

University of Maryland, College Park, MD **2021 – Present**
Ph.D. in Urban and Regional Planning and Design

- GPA: 3.83/4.0; Dean's Fellowship; Ann G. Wylie Dissertation Fellowship
- Major: Transportation Planning and Policy; Minor: Urban Data Science
- Dissertation: *"Telework and the Changing Urban Constraint: Housing, Home–Work Separation, and Uneven Commute Exposure"*; Advisor: Casey Dawkins

Ulsan National Institute of Science and Technology (UNIST), Ulsan, South Korea **2018 – 2020**
M.Sc. in Urban Infrastructure Engineering

- GPA: 4.08/4.3; Full Scholarship for two years
- Thesis: "Impact of Regional Transportation Investments on Housing Values in Central Florida"; Advisor: Gi-Hyoug Cho

Ulsan National Institute of Science and Technology (UNIST), Ulsan, South Korea **2014 – 2018**
B.Sc. in Urban Infrastructure Engineering & Business Management (Minor)

- GPA: 3.86/4.3; Full Scholarship for four years; Graduated with Honors: *Magna Cum Laude*

APPOINTMENTS

Data Science for Social Good Fellowship, Johns Hopkins University, Baltimore, MD **Summer 2026**
Summer Fellow

- Selected for a competitive 10-week fellowship with Johns Hopkins' GovEx and Baton Rouge's Department of Development to help the city address urban blight proactively rather than waiting on resident complaints.
- Built a machine-learning pipeline for early prediction of blight risk, combining spatial contagion (risk rises near blighted parcels) with temporal signals to surface at-risk properties before residents report them, so inspectors reach them first.
- Designed interventions for equity and effectiveness, directing distressed homeowners to assistance before enforcement, reserving action for absentee and investor-owned neglect, and auditing predictions to follow risk rather than complaints.

Real Estate Development, University of Maryland **Summer 2026**
Research Assistant

- Investigated the "Green Gap" in climate resilience across the Washington, D.C.-Baltimore corridor, grounded in a literature review and the hypothesis that low-income renters cluster in low-canopy, flood- and heat-exposed zones.
- Modeled whether assisted renters occupy systematically hotter, less-shaded, and more flood-prone parcels than market-rate units across the six-jurisdiction corridor.
- Synthesized results into policy recommendations for state housing agencies and developers, and am preparing the study for submission to the *Journal of Sustainable Real Estate*.

Lab @ DDOT (District Department of Transportation), Washington, DC **Summer 2025**
Research Assistant

- Developed a machine learning framework to predict citywide bike ridership using limited and uneven data, and to identify areas with unmet demand that were otherwise not observed by the agency.
- Supported data science and bike planning teams at the Office of the City Administrator and DDOT to address agency and community priorities.

National Center for Smart Growth Research, University of Maryland **Jan 2022 – May 2025**
Research Assistant

2022–2024 Annual Maryland Commuter Survey

- Led a three-year MDOT-funded study, overseeing survey design, data analysis, and technical reporting to support statewide planning for transportation infrastructure.
- Advised state officials on key findings, including that hybrid workers comprise one-third of the workforce and have the longest commutes, trip times have increased, and few auto commuters can switch to transit without substantially longer travel.

Evaluating the Maryland Commuter Tax Credit Program **Jan 2024 – Aug 2024**

- Found that under $160K of the $1M annual allocation was claimed despite legislative support.
- Conducted stakeholder interviews and analyzed program data to identify key barriers, including employer administrative burden, unclear ROI, and limited transit coverage.
- Recommended individual filing, expanded coverage eligibility, and streamlined outreach to better serve low-income and non-traditional workers.

Multi-Criteria Decision-Making Framework for Public Microtransit **Jan 2022 – Aug 2023**

- Developed service criteria using over 100 case studies and a Spatial Propensity Index to identify optimal service zones.
- Presented findings to Prince George's County and co-authored a peer-reviewed study in *Transportation Research Record*.

District Department of Transportation (DDOT), Washington, DC **Summer 2023**

Transit Operations Data Analyst Intern

- Developed a real-time Tableau visualization tool for DC Circulator operations, supported by a Python and SQL ETL pipeline.
- Standardized data entry processes to improve the consistency of operational reporting for RATP Dev transit operator personnel.

TEACHING APPOINTMENTS

URSP601 Quantitative Research Methods (Graduate Level), University of Maryland **Fall 2024, Fall 2025**

Course Instructor

- Served as sole instructor, integrating research design, quantitative methods, and applied statistics for urban policy.
- Developed curriculum in R and reproducible research, and mentored students on projects using large administrative datasets.
- Received strong evaluations for course organization, practical relevance, and support for students at all skill levels.

SELECTED PUBLICATIONS

Erdoğan, S., Nohekhan, A., **Shokputov, A.**, & Sadabadi, K. F. (2023). Multi-Criteria Decision-Making Framework for Determining Public Microtransit Service Zones. *Transportation Research Record*. DOI

MANUSCRIPTS IN PREPARATION

Shokputov, A. The Rent–Wage Scissors: Telework and the Divergence of Housing-Cost and Earnings Gradients for Place-Tied Workers in U.S. Metropolitan Areas. *In preparation*.

Shokputov, A., & Dawkins, C. Telework Intensity and Realized Home–Work Separation: Disentangling Sorting from Behavioral Change. *In preparation*.

Shokputov, A. Who Teleworks and Who Wants To? Preferences, Feasibility, and Telework Misalignment Among Workers in Teleworkable Jobs. *In preparation*.

Shokputov, A. When Hybrid Work Does Not Reduce Commuting: Stated Arrangements, Behavioral Discordance, and Weekly Commute Exposure. *In preparation*.

CONFERENCE PAPERS

Shokputov, A. (2026). Hybrid Without the Telework: Discordance Between Stated Arrangements and Commute Behavior. *ACSP*, Pittsburgh, PA.

Shokputov, A. (2026). The Rent–Wage Scissors? Telework and Divergent Accessibility Gradients for Place-Tied Workers. *Data-Intensive Research Conference* (IPUMS), Minneapolis, MN.

Shokputov, A., Harvey, C. (2025). Beyond Feasibility: Telework, Commuting, and Equity in Post-Pandemic Maryland. *ACSP*, Minneapolis, MN.

Shokputov, A., Harvey, C. (2024). Uncovering the Latent Demand for Telecommuting in Maryland. *ACSP*, Seattle, WA.

Shokputov, A., Harvey, C. (2023). Maryland's Remote Work Landscape: Commuting Behaviors and Potential VMT Reductions. *ACSP*, Chicago, IL.

Shokputov, A., Iseki, H. (2023). Key Factors Driving Plug-in Electric Vehicle Adoption. *Transportation Research Forum*, Chicago, IL.

Zou, Z., **Shokputov, A.**, & Erdogan, S. (2022). Commuting Trends during the COVID-19 Pandemic: Evidence from Maryland. *TRB Annual Meeting*, Washington, DC.

GRANTS AND AWARDS

- **MPower Early Scholars Investment Fund**, GA stipend and tuition support, University of Maryland Spring 2027
- **Data Science for Social Good Fellowship**: $14,500, Johns Hopkins University Summer 2026
- **Ann G. Wylie Dissertation Fellowship**: $15,000, University of Maryland Spring 2026
- **Dean's Fellowship**: $20,000, University of Maryland 2021–2022
- **Best Conference Paper Award**: Korea Planners Association Fall 2018

TECHNICAL TOOLKIT

Python, R; Machine Learning (scikit-learn, XGBoost); GIS (ArcGIS Pro, QGIS, PostGIS, GDAL, rasterio); Spatial R & Python (GeoPandas, OSMnx, sf, tidycensus); Visualization (Tableau, Shiny); SQL & Database (PostgreSQL, BigQuery, SQLAlchemy); Cloud & Workflow (Google Cloud, Docker, Git); Reporting & Survey (LaTeX, Qualtrics)